

02 Matrices

2.5 Geometric View of Matrix Multiplication

Linear Transformations, $Ax = y$, and Rotation, Scaling, Reflection, Shear, Projection



Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills for This Section

1

Understand the geometric meaning of matrix multiplication: it represents rotation, scaling, projection, shear, and other linear transformations, not just arithmetic

3

Master 2×2 matrices: planar rotation, reflection, shear, and projection

5

Understand 3×2 matrices: a 2D vector is mapped into 3D space, but the result still lies in a lower-dimensional subspace — “dimension lifting” is really an embedding

2

Understand the transform $\mathbf{Ax} = \mathbf{y}$: matrix $\mathbf{A}$ maps an input vector $\mathbf{x}$ to an output vector $\mathbf{y}$

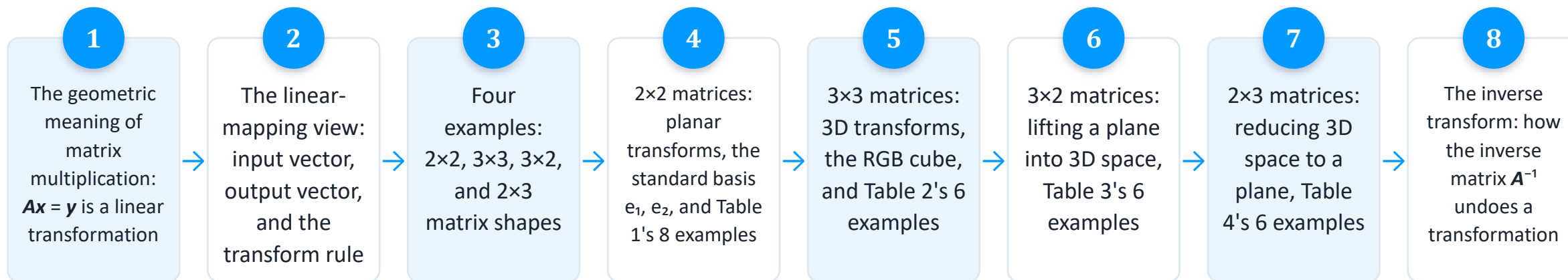
4

Master 3×3 matrices: rotation, scaling, shear, and projection in three-dimensional space

6

Understand 2×3 matrices: a 3D vector is projected down (“dimension reduction”) into a 2D plane

What This Section Covers



The Geometric Meaning of Matrix Multiplication

A matrix is a “tool” for scaling, rotating, reflecting, projecting, or shearing a vector

- Matrix multiplication is not just arithmetic on arrays of numbers — it carries rich geometric meaning
- Geometrically, matrix multiplication is fundamentally a linear transformation

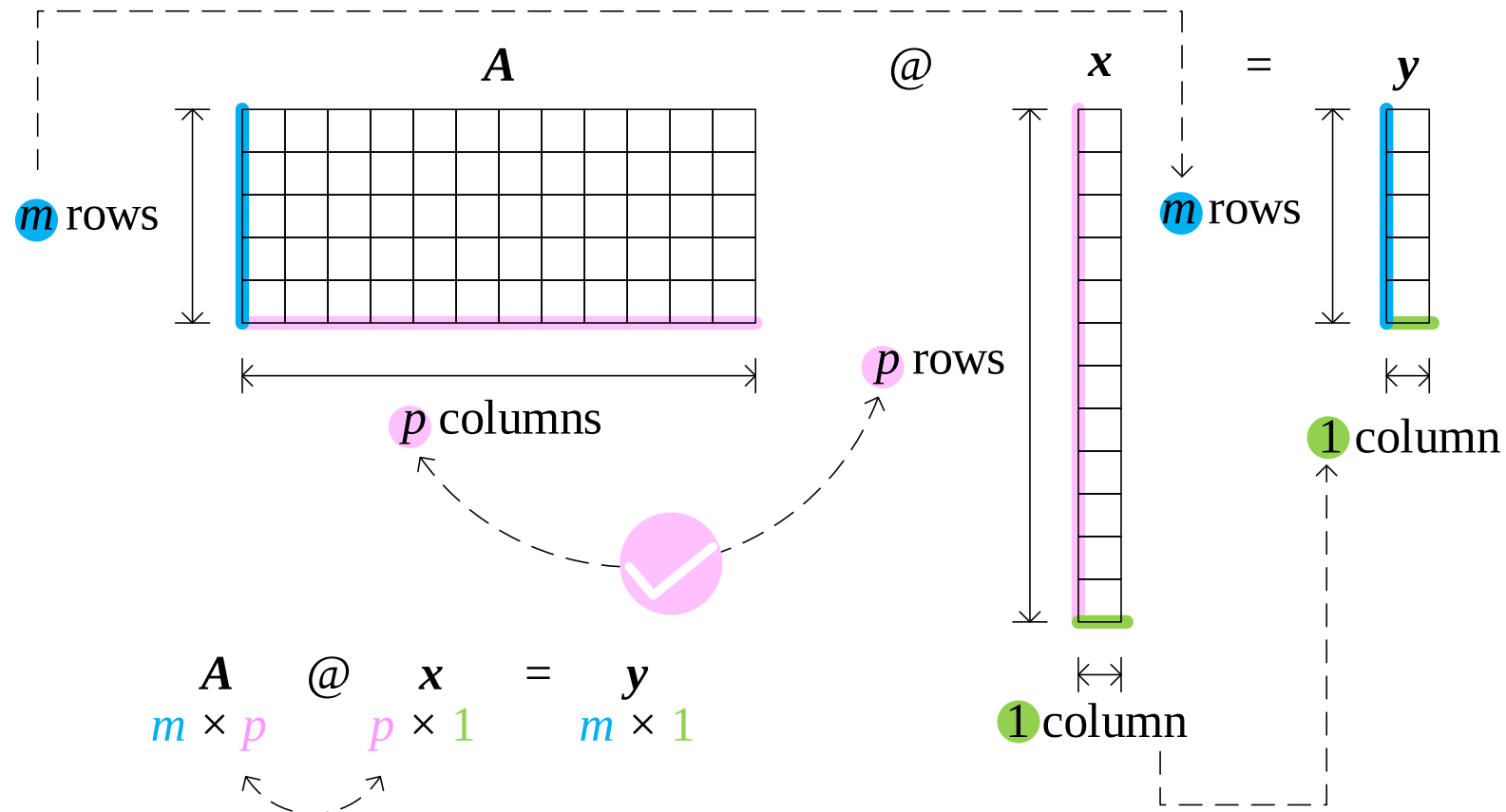
$Ax = y$: A Single Linear Transformation

$(m \times p) @ (p \times 1)$: the shared p in the middle “cancels”, leaving shape $m \times 1$

- Multiplying a vector x by a matrix A performs some geometric transformation on x
- This section builds geometric intuition for matrix multiplication from this idea

Figure 1 · Matrix Multiplication $A \mathbf{x} = \mathbf{y}$

- Matrix A acts on column vector $\mathbf{x}$ to produce column vector $\mathbf{y}$ — the basic notation used throughout this section
- The dimension of $\mathbf{x}$ is set by A 's number of columns; the dimension of $\mathbf{y}$ by A 's number of rows



The Linear-Mapping View of $A\mathbf{x} = \mathbf{y}$

$A\mathbf{x} = \mathbf{y}$ maps a p -dimensional column vector $\mathbf{x}$ to an m -dimensional column vector $\mathbf{y}$

- $\mathbf{x}$ is the input vector, an initial state — think of it as a point in space
- $\mathbf{y}$ is the output vector, the transformed result; matrix A determines the transform rule

Figure 2 · Linear Mapping $Ax = y$

- Column vector x lives in p -dimensional space, y in m -dimensional space; A maps between the two
- The transform A represents may be scaling, rotation, projection, reflection — or an ordered combination of these

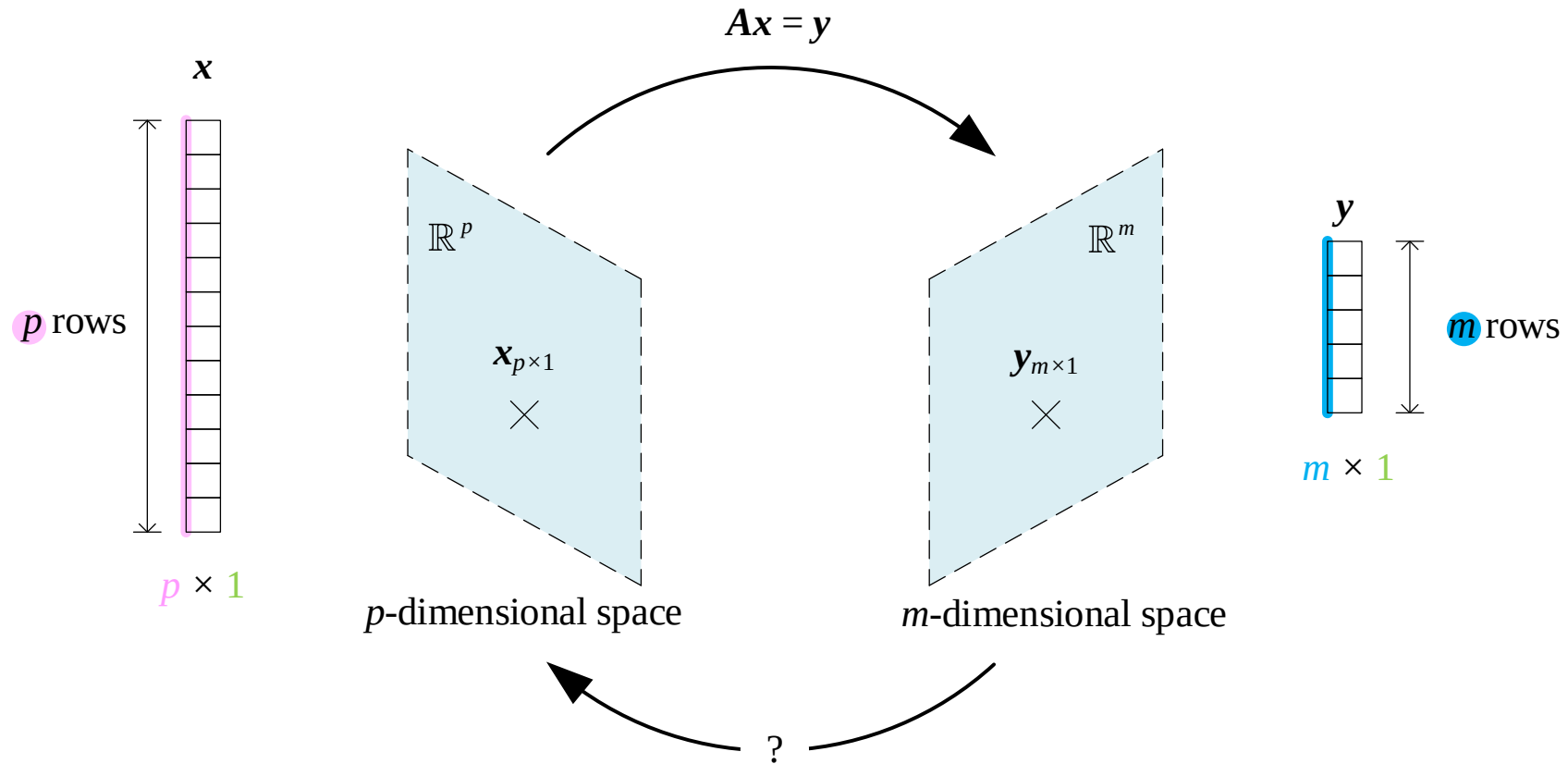
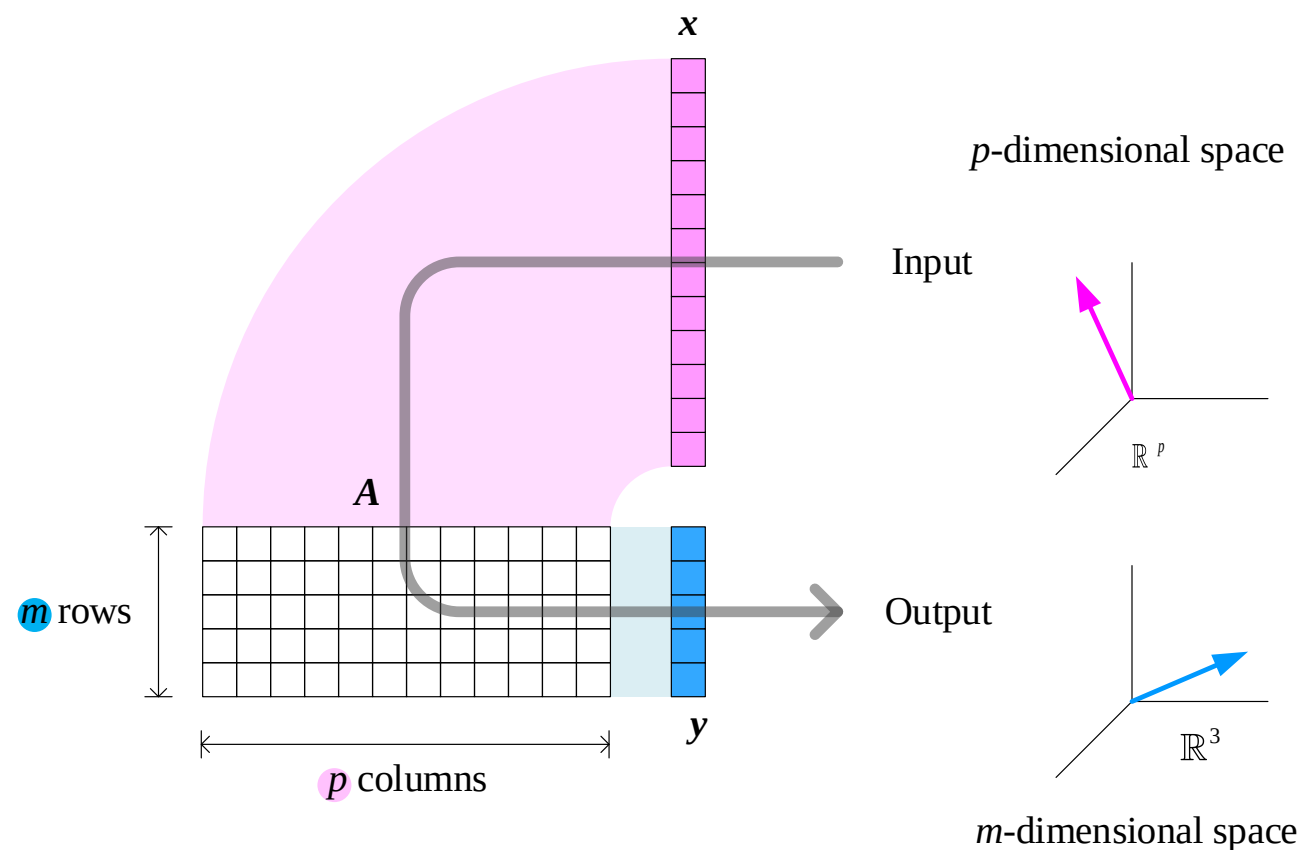


Figure 3 · Input and Output of $Ax = y$

- This layout — a column vector on the right, the matrix on the left — is the “first perspective” of matrix multiplication
- Later chapters introduce other perspectives on matrix multiplication to build a more complete picture



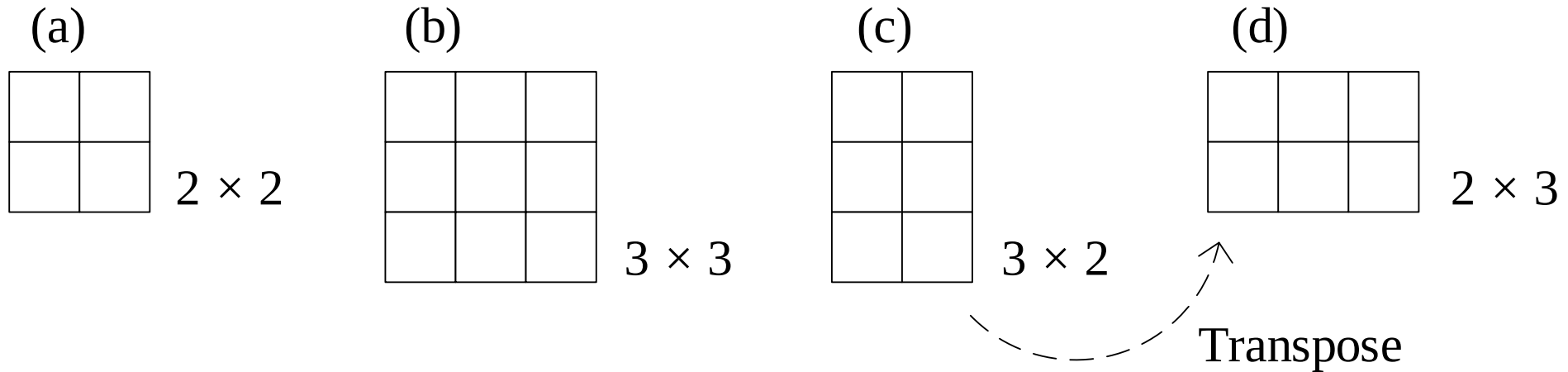
Four Examples

This section analyzes 2×2 , 3×3 , 3×2 , and 2×3 matrices and how each affects a vector

- How matrix multiplication acts geometrically on vectors is the core question this section explores
- Different matrix shapes place the input and output vectors in spaces of different dimensions

Figure 4 · Four Different Matrices A

- 2×2 : plane to plane; 3×3 : 3D space to 3D space
- 3×2 : a plane lifted into 3D space; 2×3 : 3D space reduced to a plane



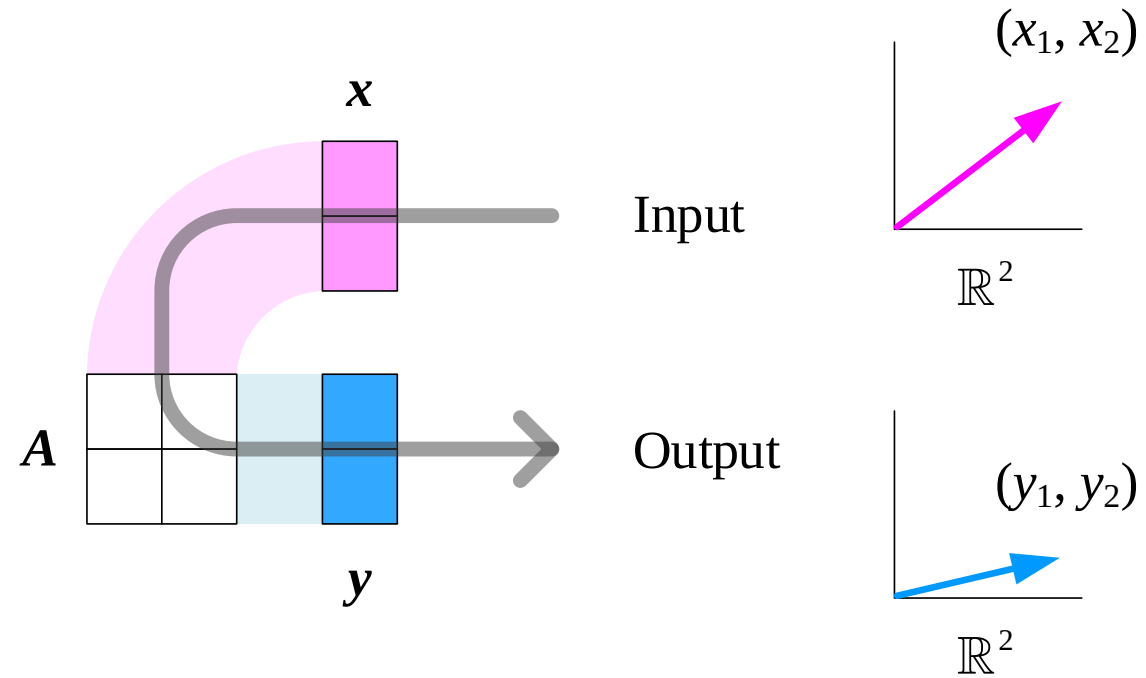
2×2 Matrices

$(2 \times 2) @ (2 \times 1)$: the shared 2 in the middle “cancels”, leaving shape 2×1

- 2×2 matrices are used for planar operations: scaling, rotation, reflection, projection, and shear
- $\mathbf{A}$'s columns must match $\mathbf{x}$'s rows; $\mathbf{A}$'s rows determine $\mathbf{y}$'s rows

Figure 5 · Matrix A is 2×2

- A 2D column vector (x_1, x_2) is both a point in the plane and a vector starting at the origin
- The output (y_1, y_2) is the same; together they form the vector space $\mathbb{R}^2$



Standard Basis Vectors e_1, e_2

$A @ e_1$ extracts column 1 of A ; $A @ e_2$ extracts column 2 of A

- e_1 points along the positive x_1 -axis, e_2 along the positive x_2 -axis — the standard basis of the plane
- On the “red-green” plane, e_1 is pure red and e_2 is pure green; the regularly spaced dots are linear combinations of e_1 and e_2

Figure 6 · Standard Basis Vectors and a Regular Grid on the “Red-Green” Plane

- The square grid is generated by linear combinations of e_1 and e_2 ; color encodes each point's “red-green” coordinates
- Next, different 2×2 matrices A will map this square into a variety of shapes

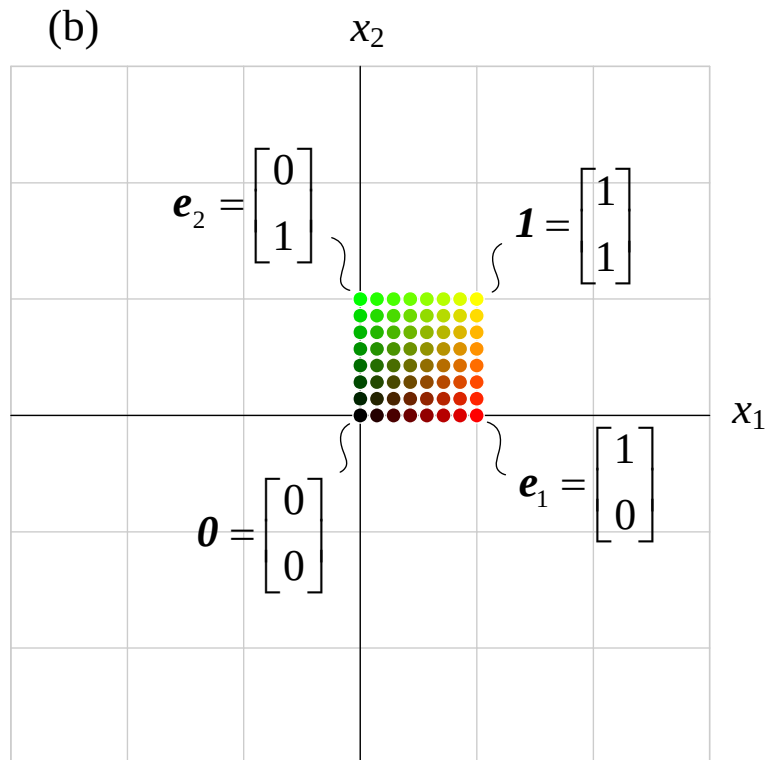
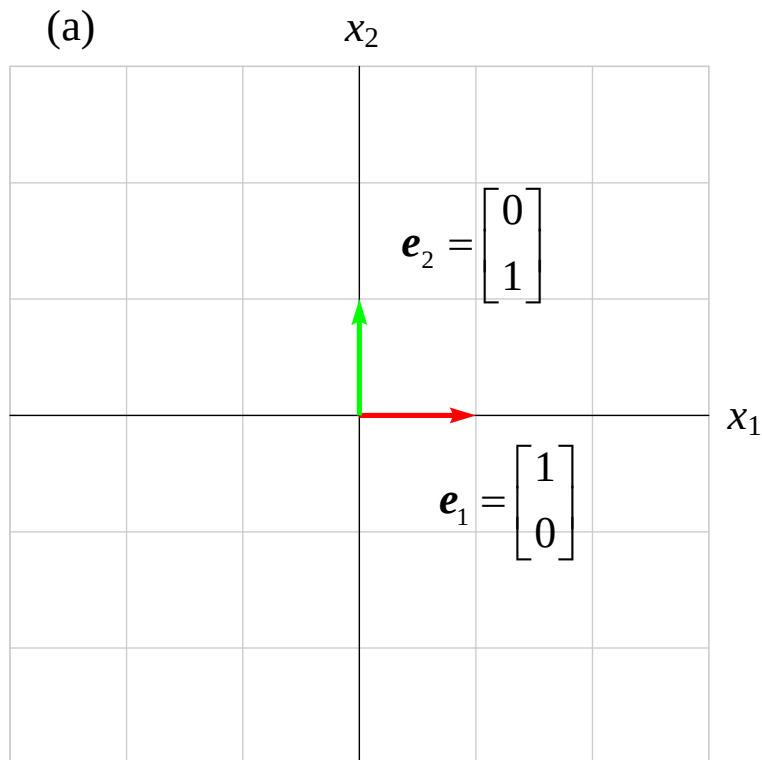


Table 1 · Analyzing 2×2 Transforms Row by Row

Under matrix A , the square grid becomes a grid of evenly spaced parallelograms

- No matter how $\mathbf{A}$ changes, the origin always stays fixed — a defining feature of linear maps
- Build intuition here; later chapters cover the underlying math of planar transforms in depth

TABLE 1 · ROW 1

Unchanged · $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

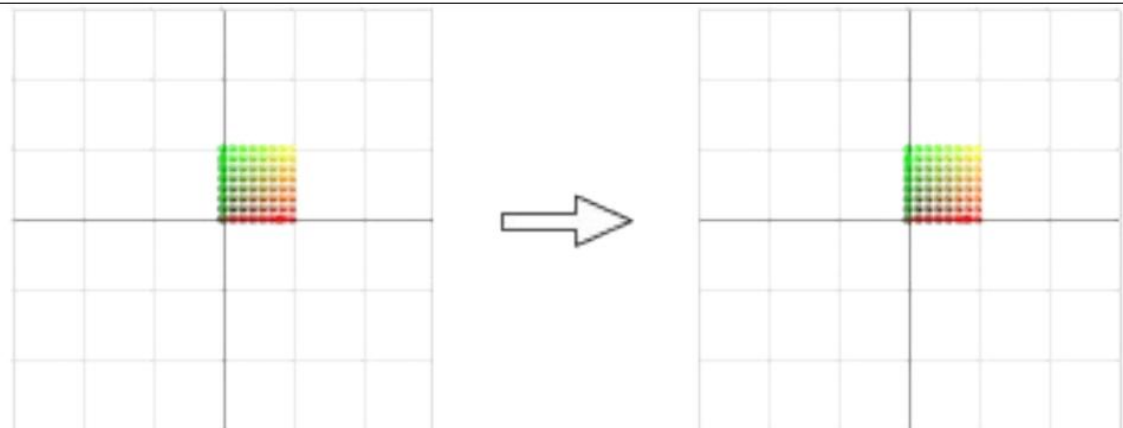
- A is the identity matrix: $A @ e_1 = e_1$ and $A @ e_2 = e_2$, so the grid is left unchanged
- The identity matrix corresponds to “doing nothing” — the most special of all linear transforms

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

不变

$$A @ e_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$A @ e_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



Uniform Stretch $\times 2 \cdot \mathbf{A} = [2, 0; 0, 2]$

- Both diagonal entries are 2, so the grid is scaled up by 2 along both x_1 and x_2
- The simplest scaling transform: the whole plane is uniformly “blown up” about the origin

$$\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

均匀拉伸 2 倍

$$\mathbf{A} @ \mathbf{e}_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\mathbf{A} @ \mathbf{e}_2 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

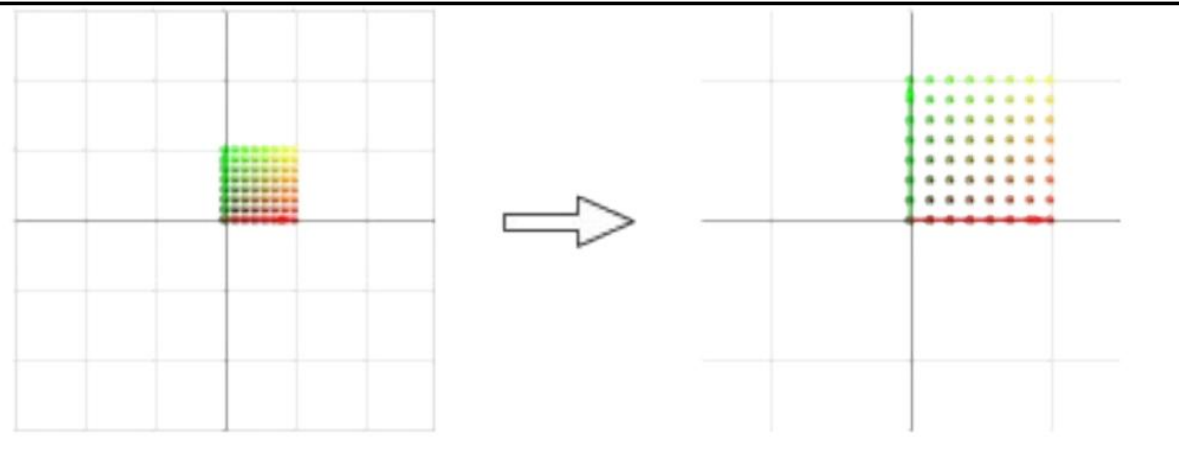


TABLE 1 · ROW 3

Squeeze · $A = [1/2, 0; 0, 2]$

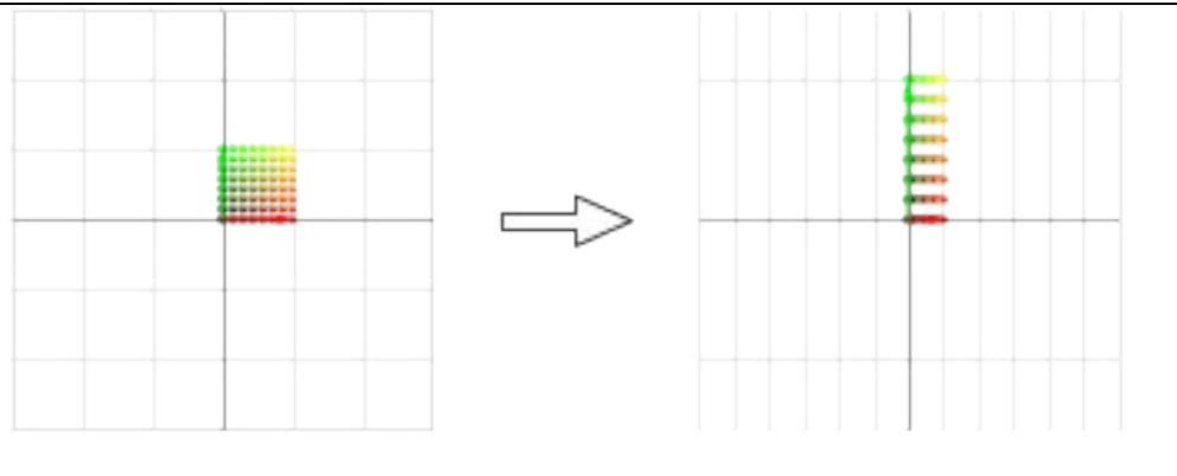
- The x_1 direction is compressed to half, the x_2 direction is stretched by 2 — the square becomes a tall rectangle
- Different scale factors along the two directions make this a non-uniform scaling

$$A = \begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix}$$

挤压

$$A @ e_1 = \begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 0 \end{bmatrix}$$

$$A @ e_2 = \begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$



Reflection about the x_1 -axis · $A = [1, 0; 0, -1]$

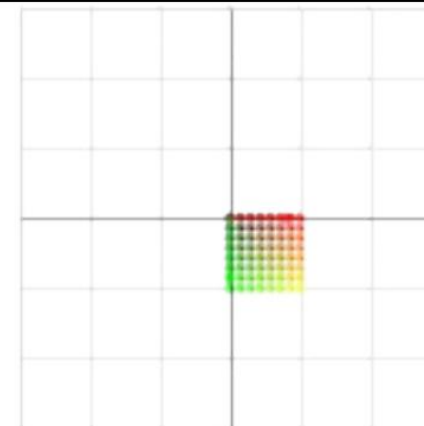
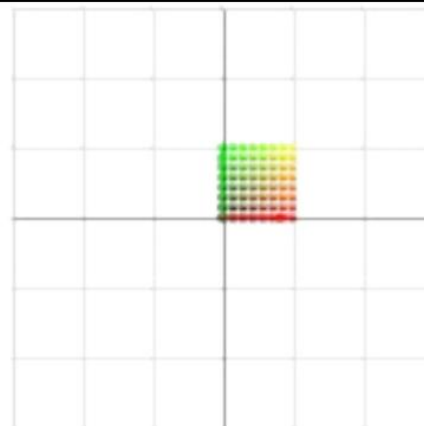
- The x_2 coordinate is negated while x_1 stays the same — the shape is mirrored across the x_1 -axis
- A reflection has negative determinant, which flips the shape's orientation (clockwise/counterclockwise)

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

关于 x_1 轴镜像

$$A @ e_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$A @ e_2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$



Point Reflection Through the Origin · $A = [-1, 0; 0, -1]$

- $A = -I$: every point maps to the point symmetric to it through the origin
- This is equivalent to a 180° rotation, or the composition of two reflections

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

关于原点对称翻转

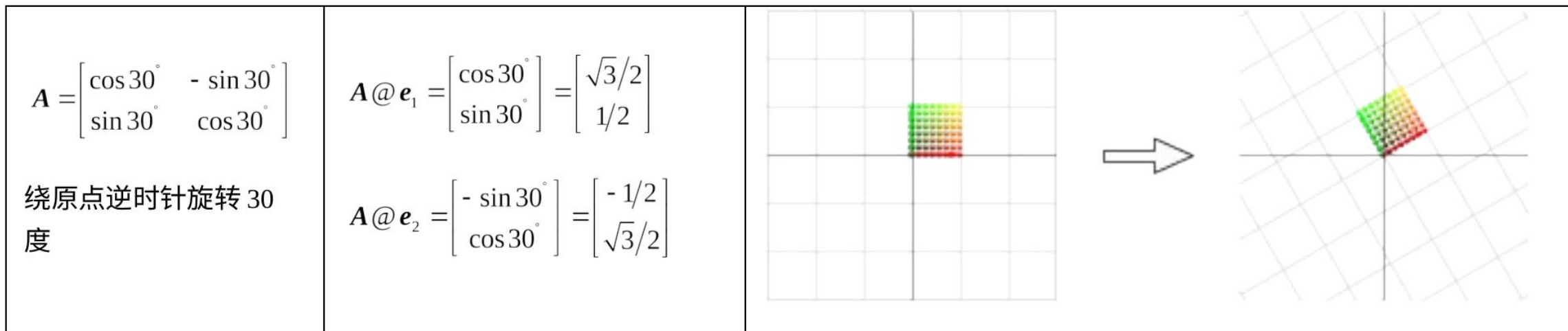
$$A @ e_1 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$A @ e_2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$



Rotate 30° Counterclockwise About the Origin

- A is built from $\cos 30^\circ$ and $\sin 30^\circ$ — the standard 2D rotation matrix
- A rotation preserves shape and size, changing only orientation



Sheared Along x_1 · $A = [1, 1; 0, 1]$

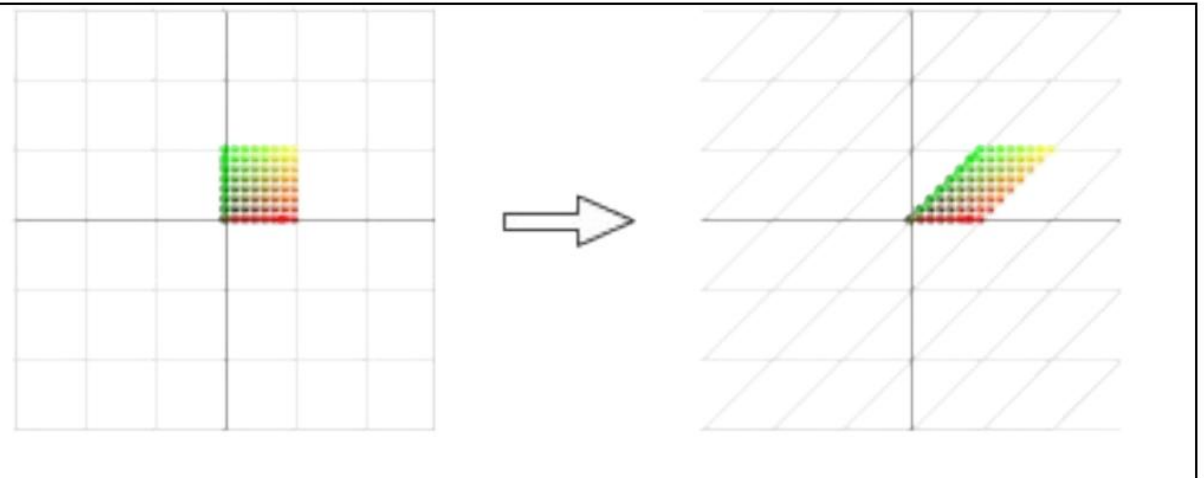
- e_2 maps to $(1,1)$ while e_1 stays fixed, pushing the square into a parallelogram
- Shear preserves area but changes the angles of the shape

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

在 x_1 方向上被剪切

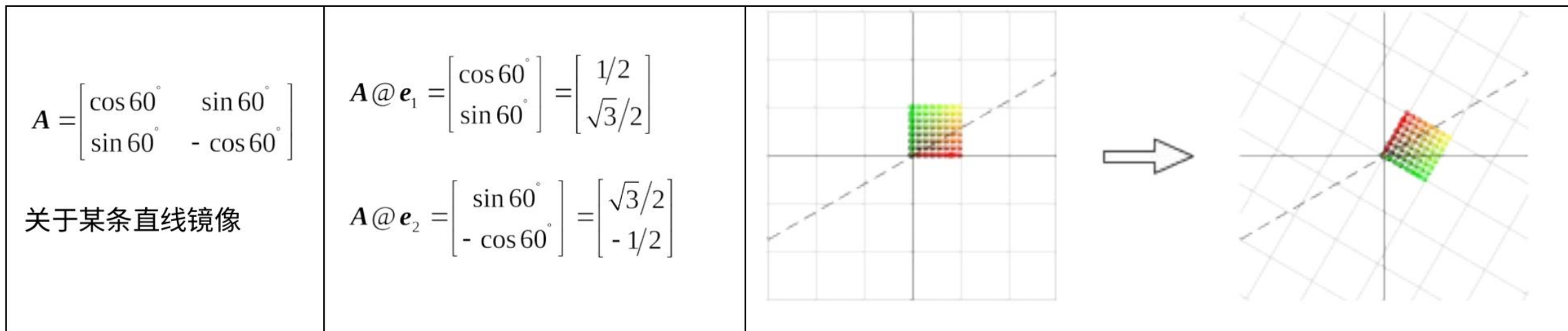
$$A @ e_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$A @ e_2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



Reflection About a Line

- $\mathbf{A}$ is built from $\cos 60^\circ$ and $\sin 60^\circ$ — the reflection matrix for a line through the origin at half that angle
- The mirror line's direction is set by the angle parameter — a “reflected version” of the rotation matrix



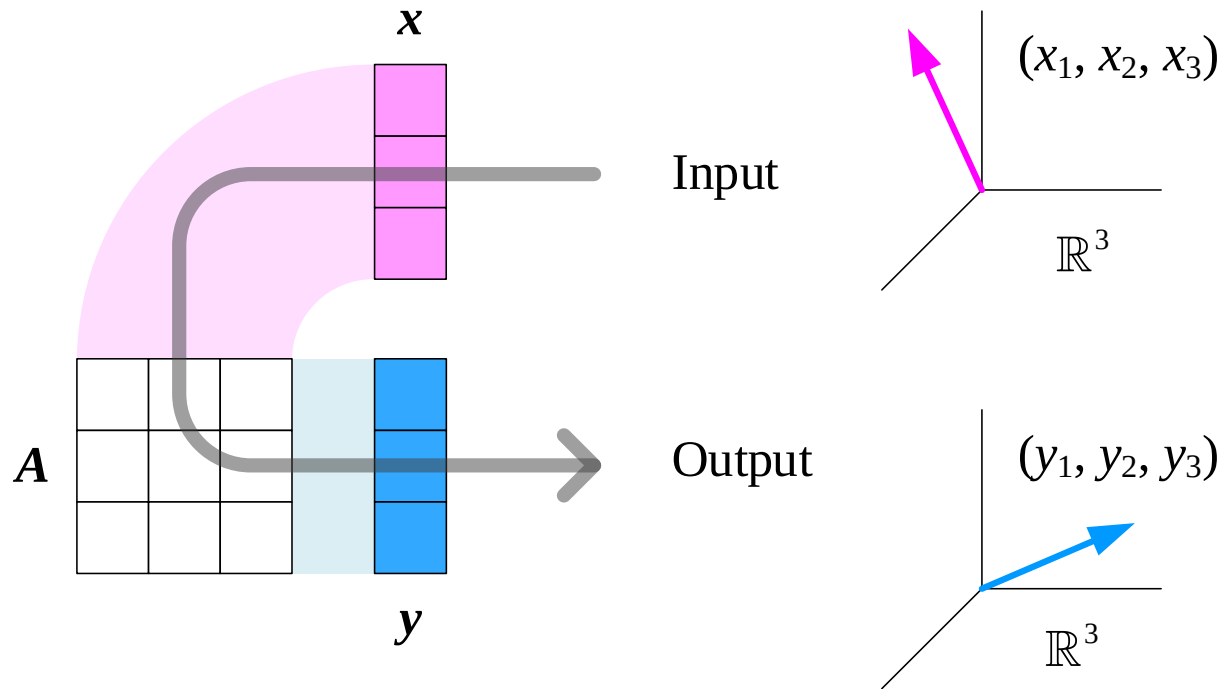
3 × 3 Matrices

$(3 \times 3) @ (3 \times 1)$: the shared 3 in the middle “cancels”, leaving shape 3×1

- 3×3 matrices extend planar transforms into 3D space, handling more complex spatial transforms
- Both the input $\mathbf{x}$ and output $\mathbf{y}$ are points in three-dimensional space

Figure 7 · Matrix A is 3×3

- A 3D column vector (x_1, x_2, x_3) is both a point in space and a vector starting at the origin
- The output (y_1, y_2, y_3) is the same; together they form the vector space $\mathbb{R}^3$



Standard Basis Vectors e_1, e_2, e_3

$e_1 = [1, 0, 0]^T$ pure red, $e_2 = [0, 1, 0]^T$ pure green, $e_3 = [0, 0, 1]^T$ pure blue

- In RGB color space, e_1, e_2, e_3 correspond to the red, green, and blue coordinate directions
- The regularly spaced dots along the RGB cube's twelve edges are linear combinations of e_1, e_2, e_3

Figure 8 · Standard Basis Vectors and Regular Dots in “Red-Green-Blue” Space

- The RGB cube's edges are spanned by e_1, e_2, e_3 ; each dot's color reflects its coordinates along the three axes
- Next, different 3×3 matrices A will map this cube into a variety of spatial shapes

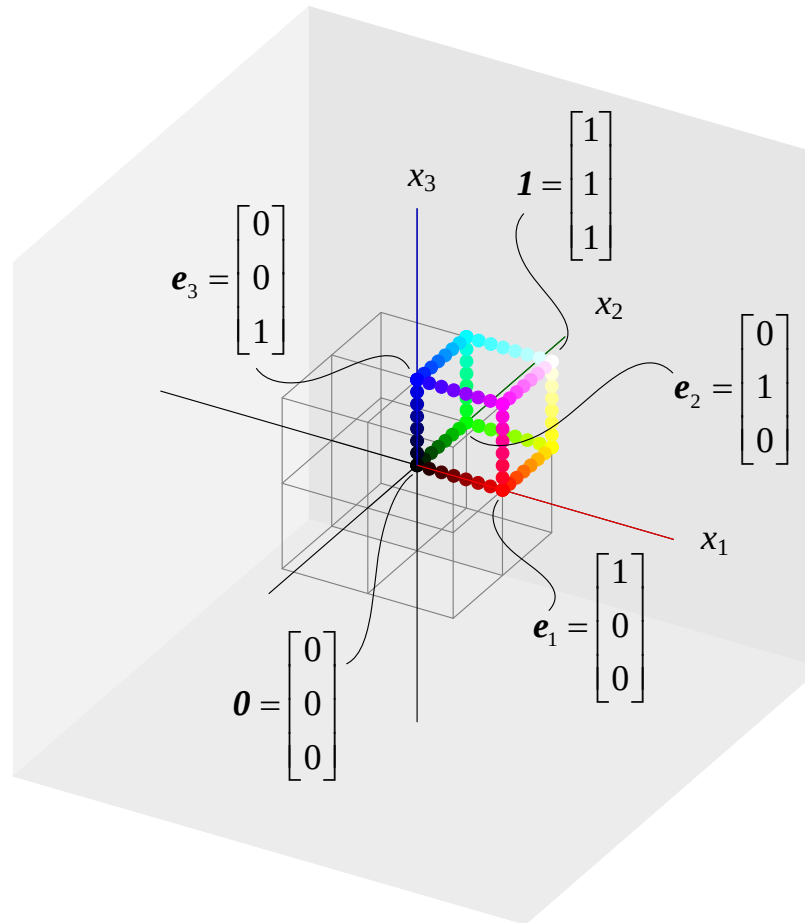


Table 2 · Analyzing 3×3 Transforms Row by Row

3×3 matrices can perform scaling, rotation, reflection, and shear in 3D space

- Following the same approach as Table 1, work through each row of Table 2 on your own
- 3D transforms are more abstract than 2D ones — try tracking how the cube's edges and corners move

Unchanged · $A = I$ (3×3 Identity Matrix)

- A is the 3×3 identity matrix: $Ae_1 = e_1$, $Ae_2 = e_2$, $Ae_3 = e_3$ — the cube is unchanged
- The identity matrix in 3D also corresponds to “doing nothing”

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

不变

$$Ae_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad Ae_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad Ae_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

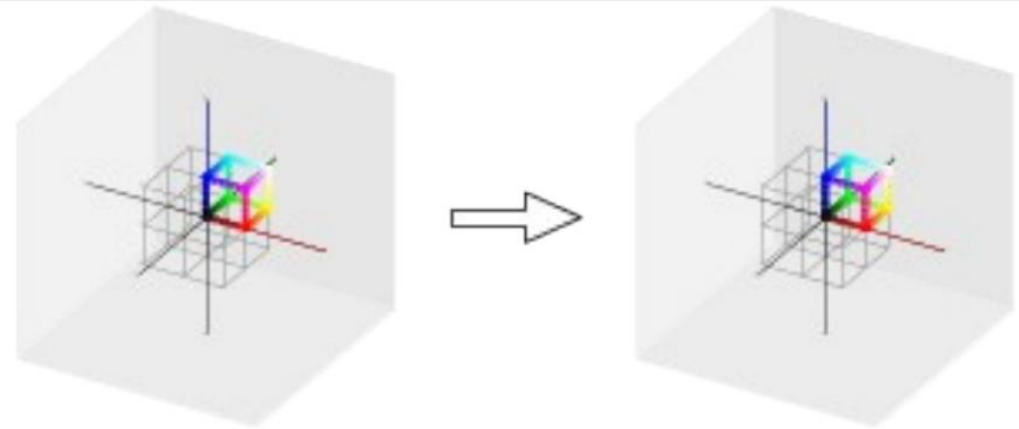


TABLE 2 · ROW 2

Uniform Stretch $\times 2 \cdot \mathbf{A} = [2,0,0; 0,2,0; 0,0,2]$

- All diagonal entries are 2, so the cube scales up by 2 along x_1 , x_2 , and x_3 at once
- A uniform 3D scaling about the origin: the cube stays a cube, just larger

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

均匀拉伸 2 倍

$$\mathbf{A}\mathbf{e}_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{A}\mathbf{e}_2 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{A}\mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

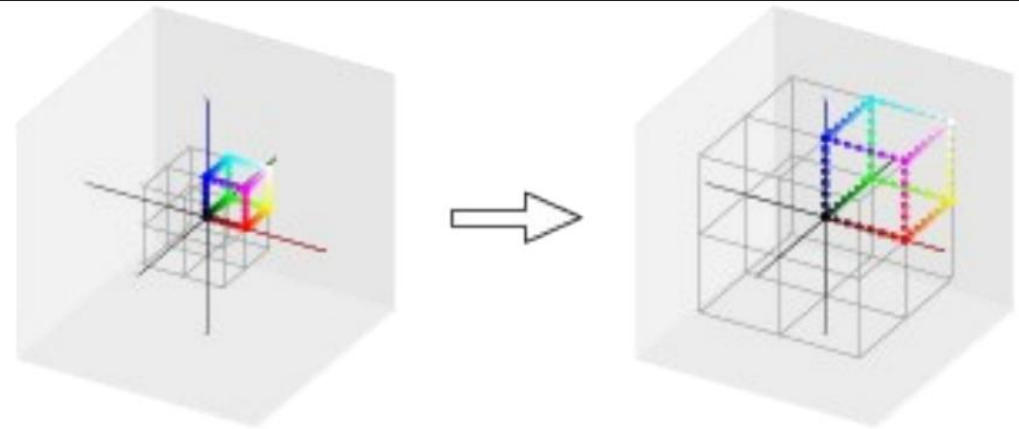


TABLE 2 · ROW 3

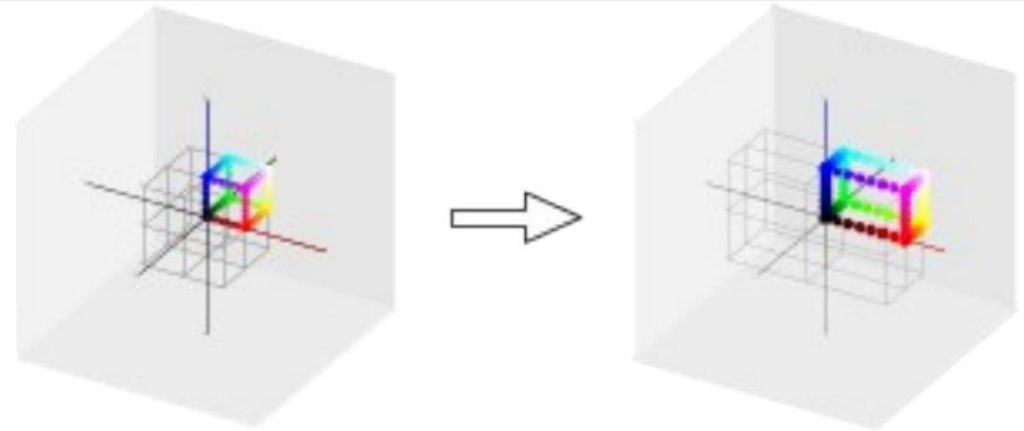
Non-Uniform Scaling · $A = [2, 0, 0; 0, 0.8, 0; 0, 0, 1.25]$

- The three scale factors — 2, 0.8, and 1.25 — are all different
- The cube becomes a rectangular box: longer along x_1 , shorter along x_2 , slightly longer along x_3

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0.8 & 0 \\ 0 & 0 & 1.25 \end{bmatrix}$$

非均匀缩放

$$A\mathbf{e}_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \quad A\mathbf{e}_2 = \begin{bmatrix} 0 \\ 0.8 \\ 0 \end{bmatrix}, \quad A\mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1.25 \end{bmatrix}$$



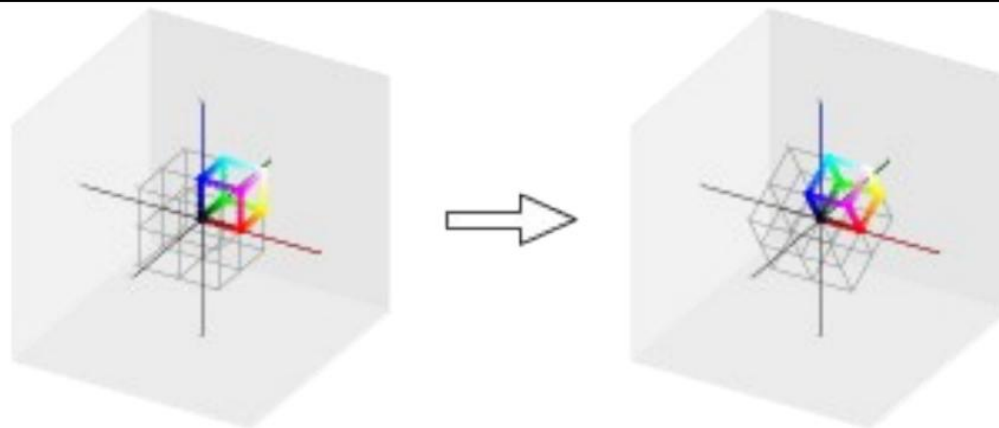
Rotation About the x_1 -axis

- A 's first row and column stay $[1,0,0]$; the x_2x_3 -plane coordinates rotate using $\cos 30^\circ$ and $\sin 30^\circ$
- Points on the rotation axis stay fixed; every other point traces a circle around that axis

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 30^\circ & -\sin 30^\circ \\ 0 & \sin 30^\circ & \cos 30^\circ \end{bmatrix}$$

绕 x_1 轴旋转

$$Ae_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad Ae_2 = \begin{bmatrix} 0 \\ \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix}, \quad Ae_3 = \begin{bmatrix} 0 \\ -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}$$



Reflection About the x_1x_2 -plane · $A = [1,0,0; 0,1,0; 0,0,-1]$

- The x_3 coordinate is negated while x_1, x_2 stay the same — a mirror reflection across the x_1x_2 -plane
- 3D reflection generalizes the 2D case directly: the mirror becomes a plane instead of a line

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

关于 x_1x_2 平面镜像

$$A\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad A\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad A\mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

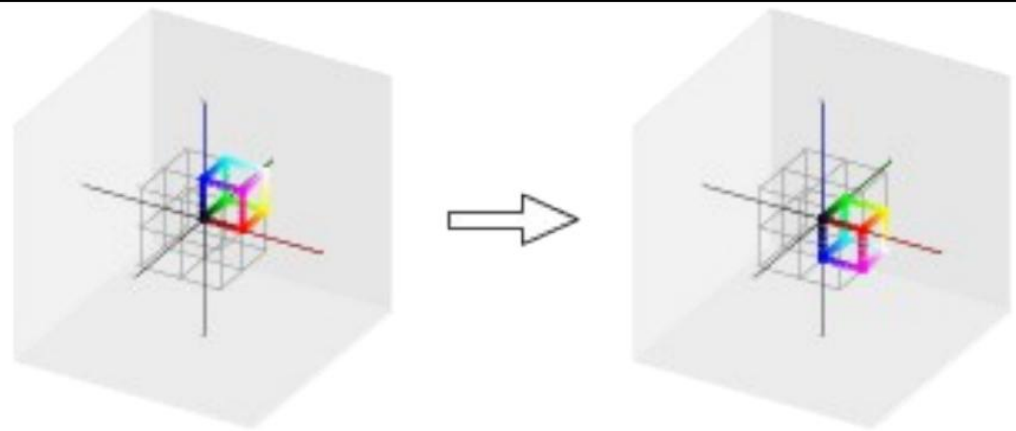
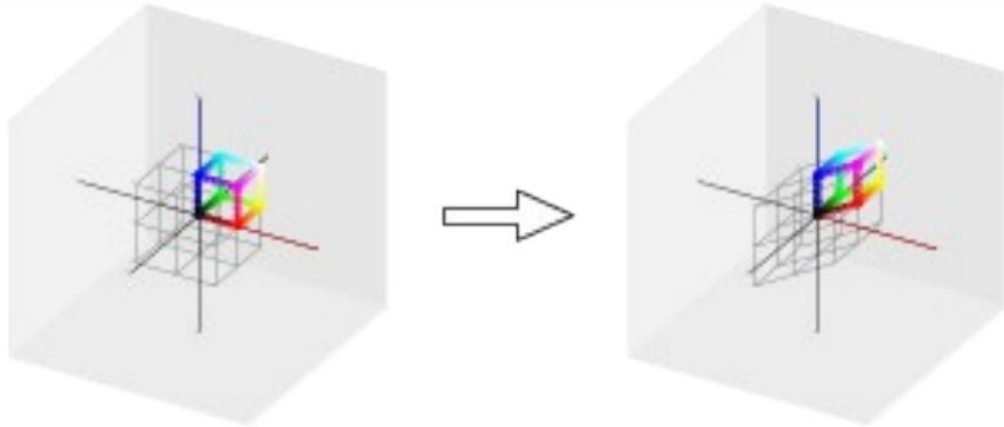


TABLE 2 · ROW 6

Shear Along x_3 · $A = [1, 0, 0; 0, 1, 0; \tan 30^\circ, 0, 1]$

- e_1 maps to $(1, 0, \tan 30^\circ)$ while e_2 and e_3 stay fixed — the cube gets “pushed” at an angle
- 3D shear also preserves volume, only changing the angles between the cube's edges

$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \tan 30^\circ & 0 & 1 \end{bmatrix}$ 沿 x_3 轴剪切	$Ae_1 = \begin{bmatrix} 1 \\ 0 \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \quad Ae_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad Ae_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	
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3×2 Matrices

$(3 \times 2) @ (2 \times 1)$: the shared 2 in the middle “cancels”, leaving shape 3×1

- 3×2 and 2×3 matrices convert between dimensions — for example, embedding a planar vector into 3D space
- The input $\mathbf{x}$ is 2D (a point in the plane); the output $\mathbf{y}$ is 3D (a point in space)

Figure 9 · Matrix A is 3×2

- The input (x_1, x_2) is a point in $\mathbb{R}^2$; the output (y_1, y_2, y_3) is a point in $\mathbb{R}^3$
- Next, different 3×2 matrices A will “embed” a square from the plane into 3D space

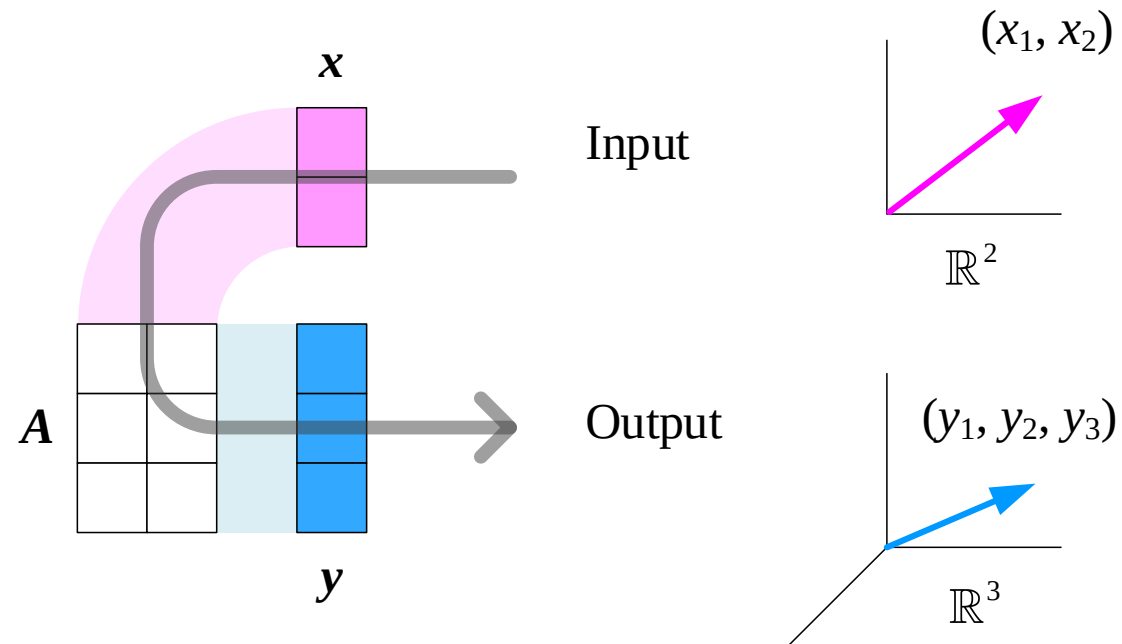


Table 3 · The “Dimension-Lifting” Transforms of 3×2 Matrices

The column space of a 3×2 matrix A has a maximum dimension of a 2D plane

- Coordinates expand from 2D to 3D, but the points don't fill all of 3D space — they still lie on a plane, so “lifting” is really an embedding
- Special cases: if $A = [1,2; 1,2; 0,0]$, the columns are linearly dependent and the column space degenerates to a line (1D); if A is the zero matrix, it degenerates to the origin (0D)

TABLE 3 · ROW 1

Lifting · $A = [1,0; 0,1; 0,0]$

- The square in the plane is embedded unchanged into the x_1x_2 -plane of 3D space (the third coordinate is always 0)
- The simplest lifting example: no scaling, no rotation — just an added coordinate that is always 0

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

升维

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

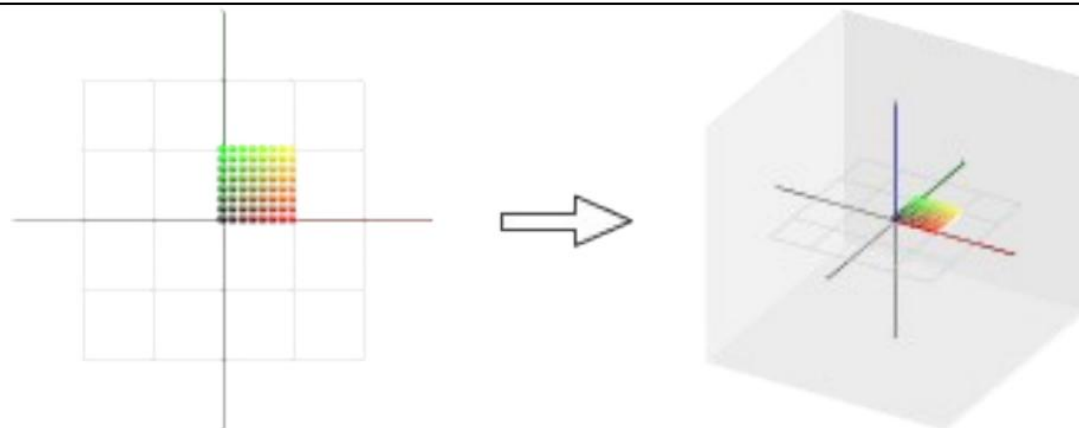


TABLE 3 · ROW 2

Scale + Lift · $A = [1,0; 0,0.5; 0,0]$

- The x_2 direction is first compressed to half, then embedded into the x_1x_2 -plane of 3D space
- Lifting can be freely combined with in-plane scaling

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0.5 \\ 0 & 0 \end{bmatrix}$$

缩放 + 升维

$$\begin{bmatrix} 1 & 0 \\ 0 & 0.5 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0.5x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

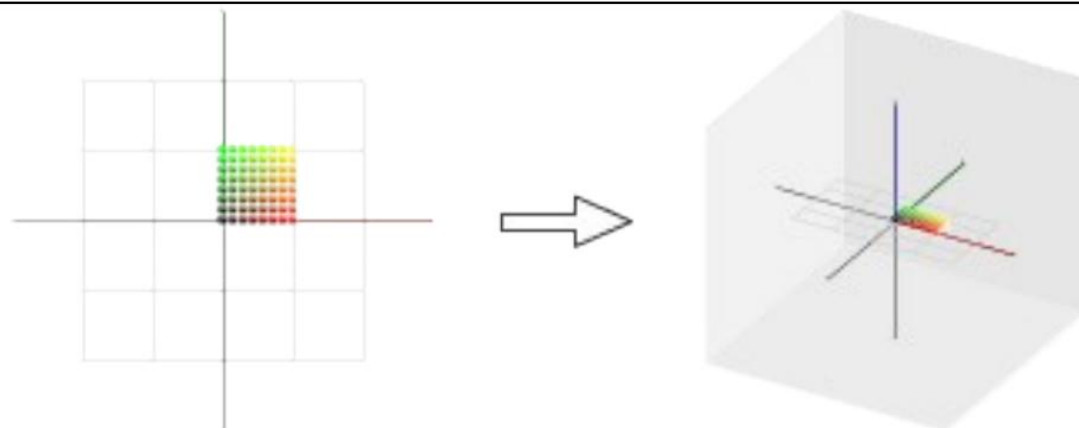


TABLE 3 · ROW 3

Lift Into the x_1x_3 -plane · $A = [1,0; 0,0; 0,1]$

- The second output coordinate is always 0, so the square is embedded into the x_1x_3 -plane instead
- The target plane of the embedding is set by A 's columns — it need not be the x_1x_2 -plane

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$$

升维到 x_1x_3 平面

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \\ x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$



TABLE 3 · ROW 4

Point Reflection + Lift · $A = [0,0; -1,0; 0,-1]$

- The planar shape first undergoes a point reflection through the origin, then is embedded into a plane in 3D
- Any planar transform applied before lifting — reflection, rotation, scaling — carries through

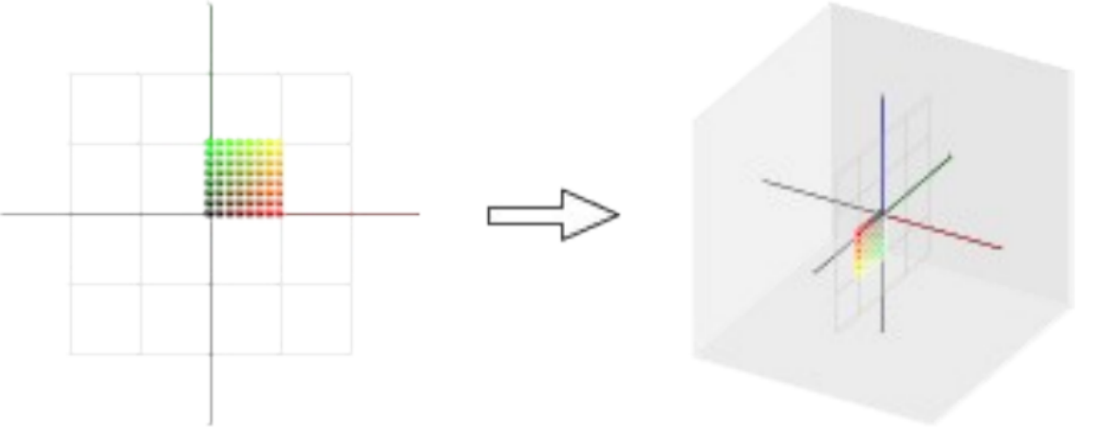
$A = \begin{bmatrix} 0 & 0 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}$ 中心对称 + 升维	$\begin{bmatrix} 0 & 0 \\ -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -x_1 \\ -x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$	
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TABLE 3 · ROW 5

Rotate + Lift · $A = [1, 0; 0, \cos 60^\circ; 0, \sin 60^\circ]$

- e_2 tilts into 3D space after the transform, so the square is embedded into a tilted plane
- This shows a 3×2 matrix can embed into a plane of any orientation in 3D space

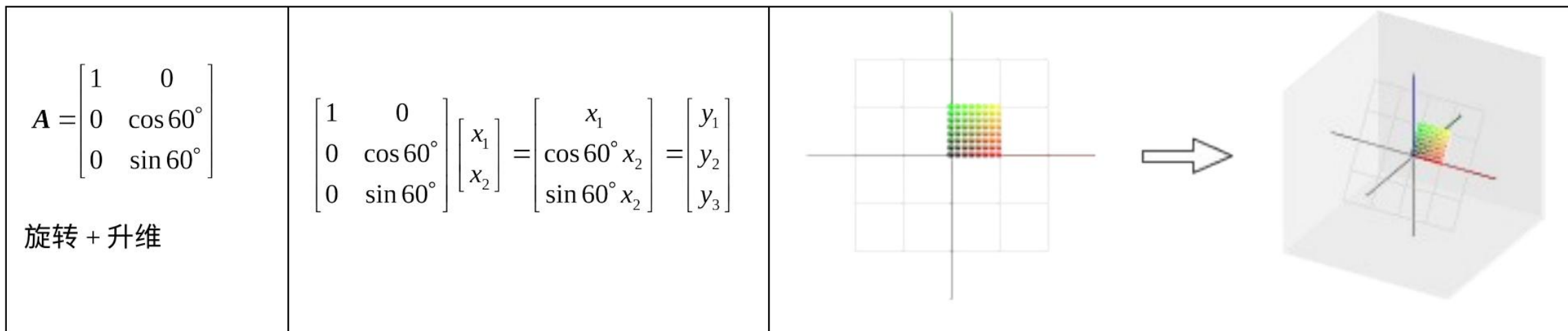


TABLE 3 · ROW 6

Shear + Lift · $A = [1,0; 1,1; 0,1]$

- The plane is first sheared, then the result is embedded into a parallelogram-shaped plane in 3D
- The embedded shape is still a 2D parallelogram — just “placed” inside 3D space

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$$

剪切 + 升维

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_1 + x_2 \\ x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$



2×3 Matrices

$(2 \times 3) @ (3 \times 1)$: the shared 3 in the middle “cancels”, leaving shape 2×1

- The input $\mathbf{x}$ is 3D (a point in space); the output $\mathbf{y}$ is 2D (a point in the plane)
- From the linear-mapping view, a flat 2×3 matrix $\mathbf{A}$ maps a 3D column vector to a 2D column vector

Figure 10 · Matrix A is 2×3

- The input (x_1, x_2, x_3) is a point in $\mathbb{R}^3$; the output (y_1, y_2) is a point in $\mathbb{R}^2$
- Next, different 2×3 matrices A will “project” a 3D cube down into the plane

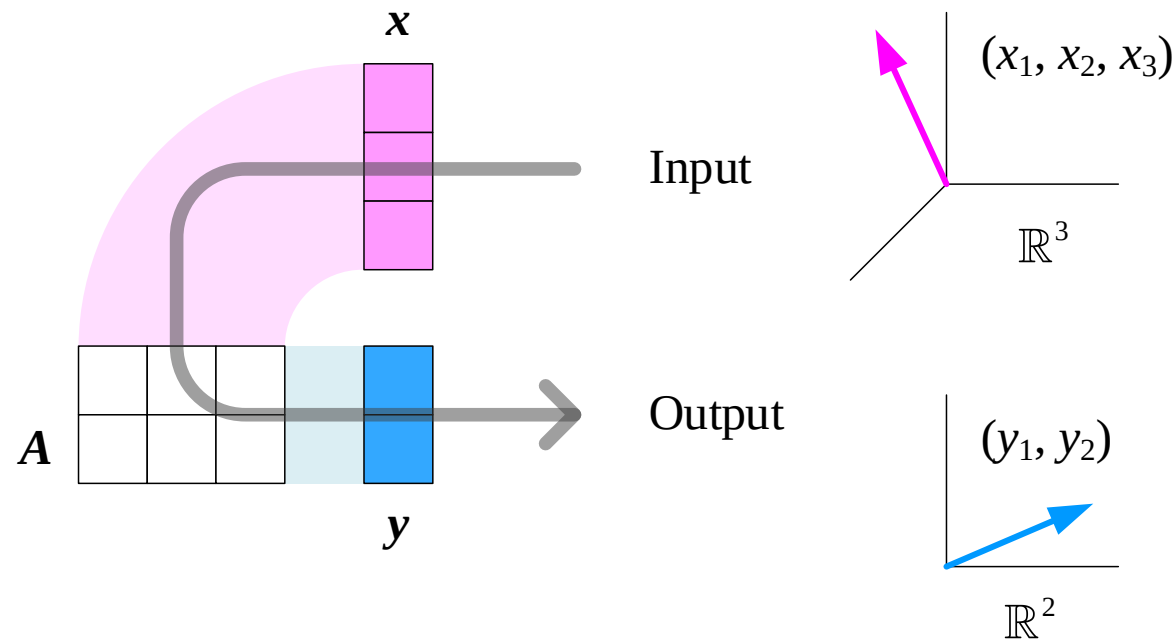


Table 4 · The “Dimension-Reduction” Transforms of 2×3 Matrices

A 2×3 matrix compresses information from 3D space into a 2D plane, keeping some linear structure

- Applying a 2×3 matrix $\mathbf{A}$ to a 3D vector $\mathbf{x}$ produces a 2D vector $\mathbf{y}$ — a projection of the 3D information onto a plane
- This “reduction” is really a projection, and it discards information along some direction

TABLE 4 · ROW 1

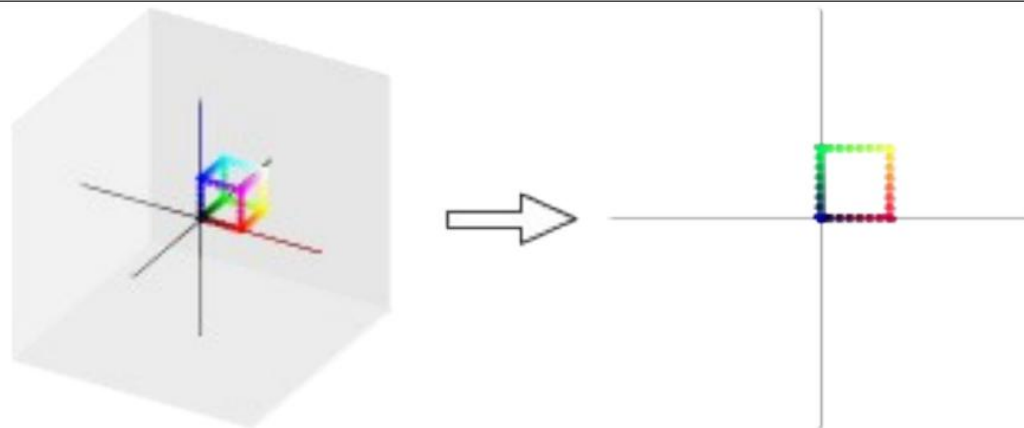
Reduction · $A = [1,0,0; 0,1,0]$

- The 3D cube's edges are projected onto the x_1x_2 -plane, simply discarding the third coordinate x_3
- The simplest reduction example: no scaling, no rotation — just a dropped coordinate

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

降维

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$



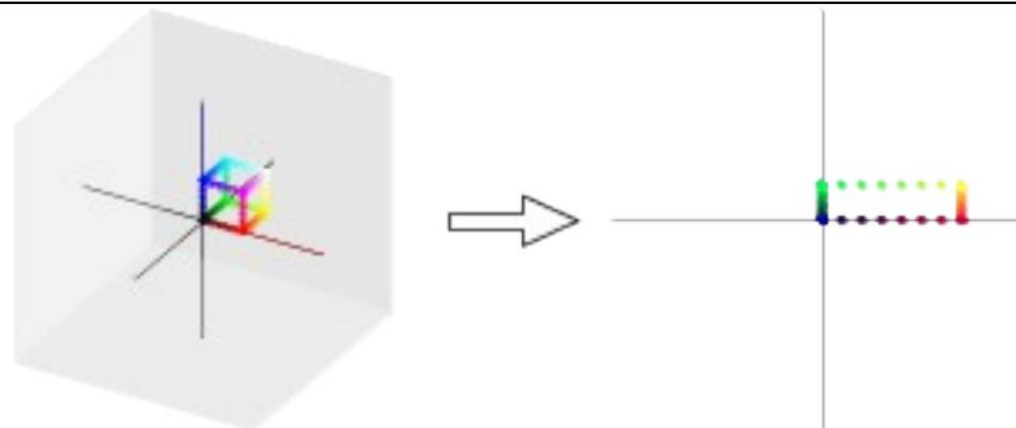
Scale + Reduce · $A = [2,0,0; 0,0.5,0]$

- After projecting to the plane, the x_1 direction is doubled and the x_2 direction is halved
- Reduction can be freely combined with in-plane scaling

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix}$$

缩放 + 降维

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_1 \\ 0.5x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$



Point Reflection + Reduce · $A = [0, -1, 0; 0, 0, -1]$

- The cube is first projected to the plane, then reflected through the origin
- The order doesn't affect the final result — projection and point reflection compose either way

$$A = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

中心对称 + 降维

$$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -x_2 \\ -x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

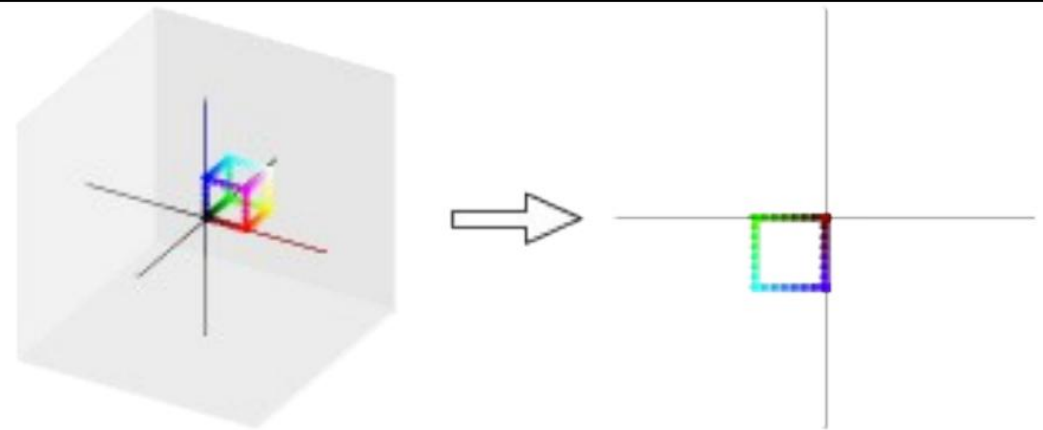
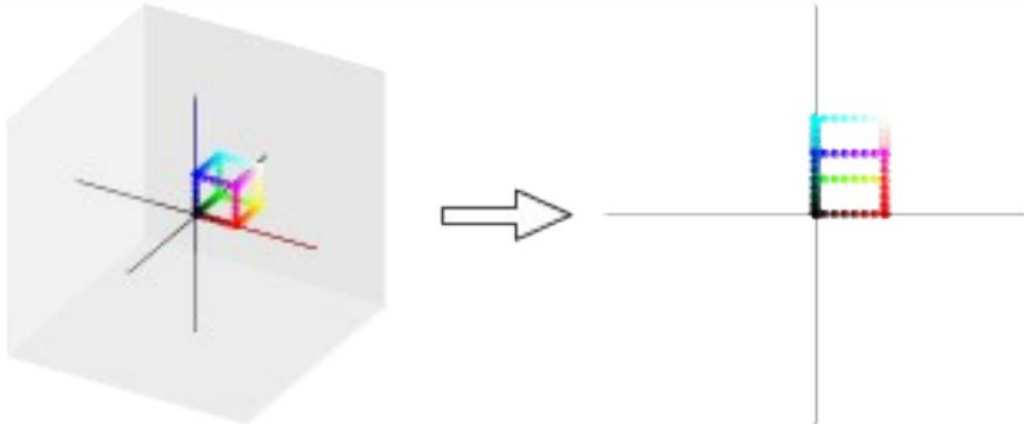


TABLE 4 · ROW 4

Rotate + Reduce · $A = [1, 0, 0; 0, \cos 60^\circ, \sin 60^\circ]$

- After projecting the cube to the plane, the resulting shape is rotated relative to the axes
- Rotation parameters can be built directly into a 2×3 matrix

$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 60^\circ & \sin 60^\circ \end{bmatrix}$ 旋转 + 降维	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 60^\circ & \sin 60^\circ \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ \cos 60^\circ x_2 + \sin 60^\circ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$	
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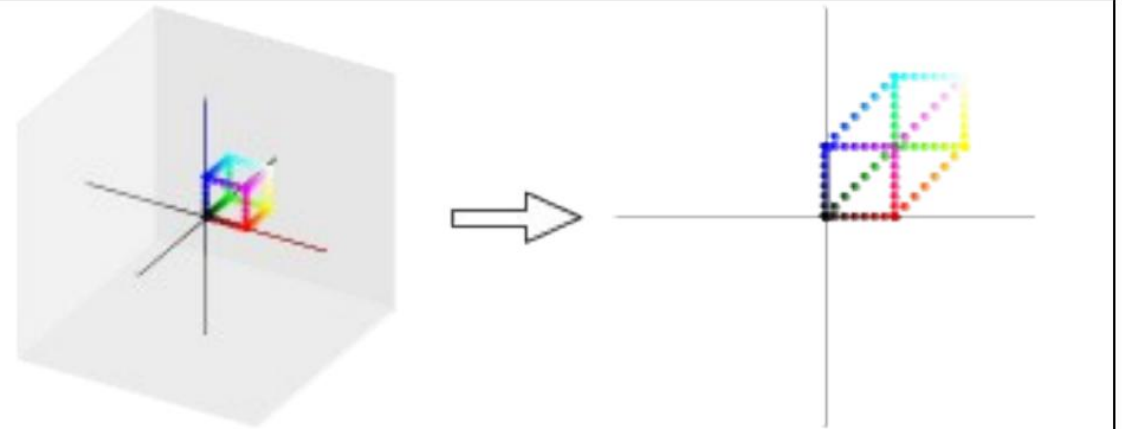
Shear + Reduce · $A = [1,1,0; 0,1,1]$

- After projecting to the plane, the square outline is pushed into a parallelogram
- Combining shear with projection means the reduced shape is no longer a plain rectangle

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

剪切 + 降维

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_2 + x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$



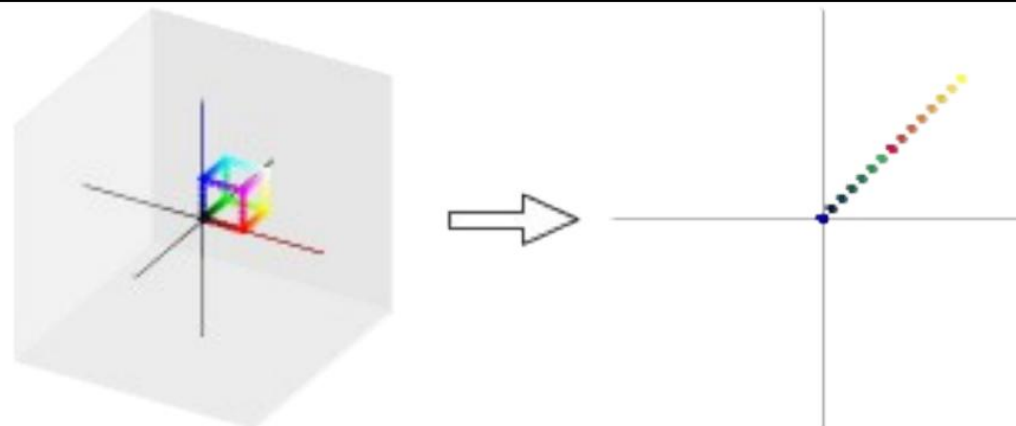
Projection + Scale + Reduce · $A = [1,1,0; 1,1,0]$

- Both rows of A are identical, so y_1 always equals y_2 — every point collapses onto a single diagonal line
- Here A 's rows are linearly dependent, so the output space degenerates further, to 1D

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

投影 + 缩放 + 降维

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_1 + x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$



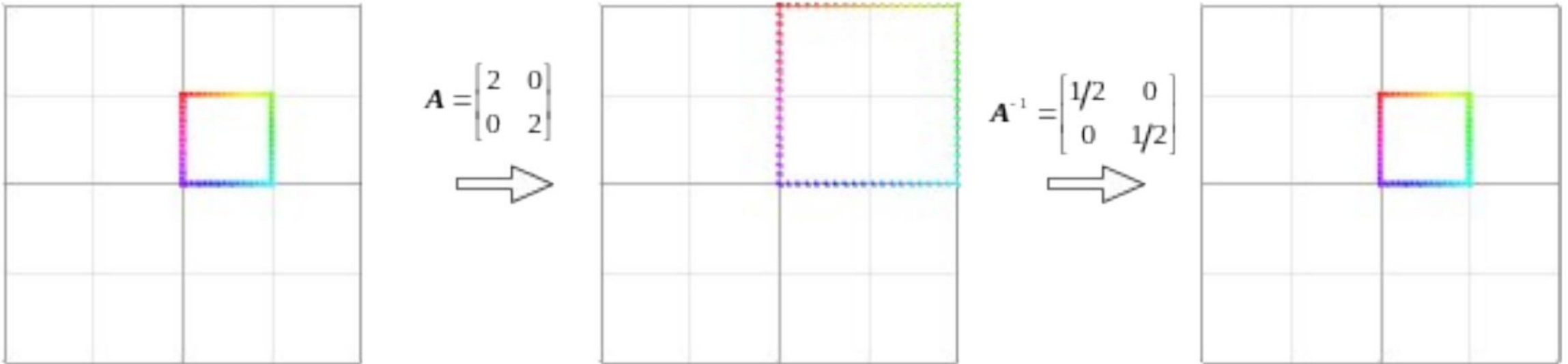
The Inverse Transform

When square matrix A is invertible, there exists A^{-1} such that $AA^{-1} = A^{-1}A = I$

- Whether y can be converted back to x depends on whether A has an inverse; only a square matrix can have one
- I is the identity matrix with the same shape as A ; A^{-1} undoes the transform performed by A

Figure 11 · Uniform Scaling in the Plane, a 2×2 Matrix

- Matrix $\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ scales the square up by 2; its inverse $\mathbf{A}^{-1} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}$ scales it back down by half
- Whether you scale up then down, or down then up, the shape is restored either way



The Geometric View Makes Matrix Multiplication Click

This section showed the rich geometric transforms matrix multiplication performs across dimensions

- Beyond matrix multiplication itself, this geometric view will also matter when studying matrix decompositions
- The next section covers the properties of matrix multiplication — this geometric intuition will make them easier to remember

Key Terms in This Section

Linear Transformation *The geometric essence of matrix multiplication*

Standard Basis *e.g. e_1, e_2, e_3 — unit vectors along the positive coordinate axes*

Rotation *Performed by an orthogonal matrix built from cos and sin*

Shear *Preserves area/volume while changing angles*

Embedding (Lifting) *Places a lower-dimensional space inside a higher one, e.g. a 3×2 matrix*

Linear Mapping *$Ax = y$ as a map from one vector space to another*

Scaling *Stretching or compressing along a given direction*

Reflection *A symmetric transform about a line or a plane*

Projection (Reduction) *Compresses higher-dimensional information into a lower dimension, e.g. a 2×3 matrix*

Matrix Inverse *Satisfies $AA^{-1} = A^{-1}A = I$; undoes the original transform*

FURTHER LEARNING

Companion Code and Further Reading

Further Reading · LA_02_05.ipynb

Code for every figure in this section — work through it line by line

Recommended · Matrix Power

A deeper look at the geometric intuition and algebra behind linear transforms

Coming Up · Rank and Column Space

Later chapters explain the math behind “lifting” and “reduction” in depth

Open-Source Repository

Code and figures for the whole book: <https://github.com/Visualize-ML>

Exercises Q1 – Q10

Q1 In the plane, how do you perform scaling with matrix multiplication?

Q3 In the plane, how do you perform rotation with matrix multiplication?

Q5 In the plane, how do you perform reflection about the x_1 -axis with matrix multiplication?

Q7 In the plane, how do you perform shear along the x_1 -axis with matrix multiplication?

Q9 Using Python, construct a 3×2 matrix A whose columns are linearly dependent, and verify its column space degenerates to 1D.

Q2 In 3D space, how do you perform rotation about the x_1 , x_2 , and x_3 axes with matrix multiplication?

Q4 In 3D space, how do you perform scaling with matrix multiplication?

Q6 In 3D space, how do you perform reflection about the x_1x_2 -plane with matrix multiplication?

Q8 In 3D space, how do you perform shear along the x_1 -axis with matrix multiplication?

Q10 Using Python, construct a 2×3 matrix A with two identical rows, and verify the output vector y always lies on a single line.

Exercises Q11 – Q20

Q11 Using NumPy and matplotlib, reproduce the rotation in Table 1 Row 6; try changing the angle from 30° to 45° and 90° , and observe the results.

Q13 Using NumPy, verify that the rotation matrices about the x_1 , x_2 , and x_3 axes all satisfy $R^T R = I$ (orthogonality).

Q15 Construct a 2×3 zero matrix, and verify that every 3D vector maps to the origin under this transform.

Q17 Write a function `apply_transform(A, points)` that applies matrix A to a set of 2D points, and use matplotlib to plot the points before and after.

Q19 Construct two 2×2 matrices — reflection about the x_1 -axis, and point reflection through the origin — and verify their product is equivalent to a single rotation.

Q12 Write a function `is_rotation(A)` that checks whether a 2×2 matrix is a rotation matrix (orthogonal with determinant 1).

Q14 Construct a 3×2 matrix A whose column space is exactly a line in 3D space, and verify this with Python.

Q16 Using Python, verify the property in Figure 11: does $A @ A_{\text{inv}}$ equal the identity matrix I (accounting for floating-point error)?

Q18 Using Python, construct a 2×2 shear matrix and verify that the shape's area is preserved before and after the shear (using the determinant).

Q20 Using Python, generate a 3×3 non-uniform scaling matrix and use a 3D scatter plot to show its effect on the RGB cube.

Exercises Q21 – Q28

Q21 Construct a 3×2 matrix A that rotates first and then “lifts”; use a 3D scatter plot to show the planar square embedded in 3D space.

Q23 Write a function `matrix_rank_via_svd(A)` and use NumPy's `np.linalg.matrix_rank()` to verify the dimension of the column/row space for matrices in Tables 3 and 4.

Q25 Using Python, construct all 6 matrices in Table 2, plot their effect on the RGB cube, and arrange the results in a 3×2 grid of subplots.

Q27 Using NumPy, verify that if A is an orthogonal matrix (rotation or reflection), applying it to any vector preserves the vector's length.

Q22 Construct a 2×3 matrix A that shears first and then “reduces”; use a 2D scatter plot to show the 3D cube projected onto the plane.

Q24 Using Python, construct all 8 matrices in Table 1, plot their effect on the square grid, and arrange the results in a 4×2 grid of subplots.

Q26 Given an arbitrary 2×2 matrix A , write a function `classify_transform(A)` that attempts to classify the transform as closer to a rotation, a scaling, or a shear.

Q28 Combining what you learned in this section, write a small program that takes a 2×2 or 3×3 matrix, prints whether it is invertible, and if so computes and displays A^{-1} .

10 Prompts to Explore Further (1/2)

- 1 Explain why any 2×2 rotation matrix can be written as $[\cos\theta, -\sin\theta; \sin\theta, \cos\theta]$, and what θ means geometrically.
- 2 Explain why the “dimension lift” produced by a 3×2 matrix is really an embedding, not a true increase in degrees of freedom.
- 3 Explain why the “dimension reduction” produced by a 2×3 matrix is really a projection, and what information it discards.
- 4 Prove that the determinant of a shear transform is always 1, and explain why this means area/volume is preserved.
- 5 Explain the difference in determinant sign between reflection and rotation, and what that sign means geometrically.

10 Prompts to Explore Further (2/2)

6

Explain why the inverse A^{-1} only exists when A is a square matrix, and whether a non-square matrix can have some kind of “pseudo-inverse”.

7

Using a geometric argument, explain why multiplying two rotation matrices is equivalent to adding their rotation angles.

8

Explain why orthogonal matrices (rotations, reflections) always preserve vector length, and connect this to $A^T A = I$.

9

Give an example of how the geometric view of matrix multiplication applies in machine learning, such as the linear layer of a neural network.

10

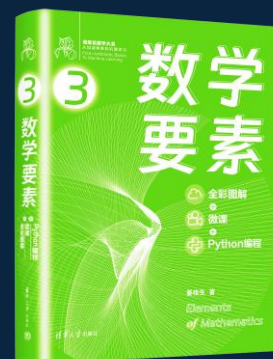
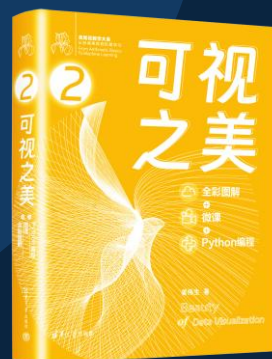
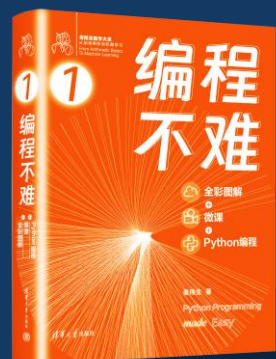
Explain why the matrix in Table 4 Row 6, which has two identical rows, causes the output dimension to degenerate further, and how this relates to the matrix's rank.

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

